

Wooldridge-Style Difference-in-Differences and the H-2A AEWR Setting

Purpose

This note explains the Wooldridge approach to modern difference-in-differences (DiD), with special attention to the role of two-way Mundlak regressions and the connection to two-way fixed effects (TWFE). It is written for readers who already understand standard DiD, event-study regressions, and fixed effects regressions, but who have not worked much with random effects models or the Mundlak-Chamberlain device.

The second half maps the approach to the current H-2A project. The current main specification is implemented in `code/c04_analysis_02_main_dd_event_study.R` and uses a continuous treatment intensity:

```
h2a_cert_share_farm_workers_2011_start_year ~  
  aewr_cz_p25_l1 * postdummy |  
  county_fe + year_fe
```

The treatment intensity is the lagged AEWR bite:

```
aewr_cz_p25_l1 = aewr_ppi_l1 - wage_p25_l1
```

That is, the real lagged regional AEWR minus the lagged CZ-level 25th percentile agricultural wage. This object is not a binary treatment indicator. It is a continuous exposure measure that varies over counties/CZs and years and is experienced by all observations to some degree.

Executive Summary

Wooldridge's recent DiD work is best understood as a regression architecture for modern DiD settings. The core idea is not that TWFE should be abandoned. Rather, the idea is that TWFE should be written in a way that makes the identifying comparisons, heterogeneity, pretrend restrictions, and aggregation weights explicit.

For binary staggered-adoption designs, this means replacing a single TWFE treatment coefficient with a richly interacted regression that estimates cohort-by-time treatment effects. For repeated cross sections, nonlinear models, and nonbinary treatments, the same principle carries through: state the conditional parallel-trends restriction, parameterize the counterfactual untreated mean flexibly enough, then estimate interpretable treatment-effect objects.

The Mundlak-Chamberlain part matters because it shows why fixed effects and appropriately augmented random effects are closely related. A Mundlak regression adds unit-level averages

of time-varying covariates to a random effects model. Doing so allows unobserved unit effects to be correlated with the covariates. In linear models, the slope on the within-unit covariate variation matches the fixed effects slope. A two-way Mundlak regression extends that logic to both unit and time heterogeneity. Wooldridge uses this equivalence to show that modern DiD can be implemented with pooled regression while still retaining the logic of fixed effects identification.

For the H-2A paper, the main payoff is conceptual and diagnostic:

1. The current model estimates one post-2011 change in the slope relating AEWR bite to H-2A usage.
2. A Wooldridge-style model would estimate a sequence of year-specific or group-specific dose-response effects rather than a single post-period slope.
3. The most important empirical question is what variation identifies the dose response: regional AEWR changes, local CZ wage changes, or residual variation from their difference.
4. Because the bite contains local wage information, a richer DiD specification improves transparency but does not by itself solve endogeneity.
5. The cleanest Wooldridge-style robustness check is likely a flexible continuous treatment event study using predetermined or baseline bite measures, plus diagnostics that show which years, regions, and dose ranges identify the results.

1. Starting Point: What Standard TWFE DiD Does

The canonical fixed effects DiD regression is:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it}$$

where:

- i indexes units;
- t indexes time;
- Y_{it} is the outcome;
- D_{it} is treatment;
- α_i are unit fixed effects;
- λ_t are time fixed effects.

When treatment is binary and turns on once, the familiar intuition is that α_i remove permanent differences across units and λ_t remove common shocks. The coefficient β is then estimated from within-unit changes in treatment, net of common time shocks.

In the simplest two-period, two-group case, this is exactly the usual DiD:

$$\beta = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$

The trouble starts when we move beyond the canonical setup:

- treatment timing differs across units;
- effects differ by cohort, calendar year, or exposure length;
- treatment intensity is continuous rather than binary;
- everyone is exposed to some degree;
- there is no clean never-treated control group;
- treatment is partly generated by local economic conditions.

In those cases, a single TWFE coefficient can be difficult to interpret. It may combine many comparisons, including comparisons across already-treated groups, and it can implicitly apply weights that are not economically meaningful.

Wooldridge’s contribution is to show that the regression framework can still be useful, but it must be written with enough structure to make the relevant treatment effects explicit.

2. Fixed Effects, Random Effects, and Why Mundlak Matters

2.1 The Fixed Effects View

Suppose the outcome is generated by:

$$Y_{it} = X_{it}\beta + a_i + u_{it}$$

where a_i is an unobserved unit-specific component. In the H-2A setting, a_i could include persistent differences in county crop mix, grower organization, baseline H-2A familiarity, climate, distance to labor supply networks, or local political economy.

The fixed effects estimator allows a_i to be arbitrarily correlated with X_{it} . This is why applied researchers like fixed effects. We do not need to assume that counties with higher AEWR bite are otherwise comparable to counties with lower AEWR bite. We only need identifying variation within counties over time after removing common shocks.

The fixed effects transformation subtracts unit averages:

$$Y_{it} - \bar{Y}_i = (X_{it} - \bar{X}_i)\beta + (u_{it} - \bar{u}_i)$$

The unobserved a_i drops out.

With two-way fixed effects, we also subtract time averages:

$$\ddot{Y}_{it} = \ddot{X}_{it}\beta + \ddot{u}_{it}$$

where the two-way residualized regressor is:

$$\ddot{X}_{it} = X_{it} - \bar{X}_i - \bar{X}_t + \bar{X}$$

in a balanced panel. This is the Frisch-Waugh-Lovell view of TWFE: the coefficient is obtained by regressing the residualized outcome on the residualized regressor.

2.2 The Random Effects View

A basic random effects model also writes:

$$Y_{it} = X_{it}\beta + a_i + u_{it}$$

But it usually imposes the key assumption:

$$E(a_i | X_{i1}, X_{i2}, \dots, X_{iT}) = 0$$

In words, the unobserved unit effect is assumed to be uncorrelated with the entire history of observed covariates. This is strong. In many policy settings, it is implausible.

For this project, a naive random effects assumption would say that persistent county-level unobservables are unrelated to the county's AEWR bite history. That would be hard to defend. Counties with systematically low agricultural wages, different crop structures, or stronger H-2A employer networks are likely to have both different bite values and different outcome trajectories.

So why discuss random effects at all? Because Mundlak showed that we can modify random effects so that it behaves like fixed effects for the slope coefficients we care about.

2.3 The Mundlak Device

Mundlak's idea is simple. If the unobserved unit effect a_i is correlated with the covariates, model that correlation directly using the unit-level average of the covariates:

$$a_i = \bar{X}_i\pi + r_i$$

where r_i is the remaining unobserved component that is uncorrelated with the covariate history after conditioning on $\bar{X}_i$.

Substitute this into the outcome equation:

$$Y_{it} = X_{it}\beta + \bar{X}_i\pi + r_i + u_{it}$$

This is a correlated random effects model. It is also called a Mundlak model. The crucial feature is that we include both:

- the time-varying covariate X_{it} ;
- the unit average $\bar{X}_i$.

The coefficient on X_{it} is then identified from within-unit deviations from the unit mean, just as in fixed effects. In a linear model, the Mundlak slope on X_{it} is numerically equivalent to the fixed effects slope under standard conditions.

The unit mean $\bar{X}_i$ absorbs the between-unit relationship between treatment exposure and persistent unobservables. The within-unit coefficient β is then not contaminated by the fact that high-exposure units differ permanently from low-exposure units.

2.4 Mundlak vs Chamberlain

The Mundlak specification uses the unit average $\bar{X}_i$. Chamberlain's correlated random effects formulation is more general. It lets the unobserved effect depend on the full covariate history:

$$a_i = \pi_1 X_{i1} + \pi_2 X_{i2} + \dots + \pi_T X_{iT} + r_i$$

The Mundlak version is a parsimonious restriction of this idea. Instead of including every time-specific covariate value, it includes the average.

For applied DiD, the important lesson is not that one must estimate a literal random effects model. The lesson is that fixed effects can be understood as a regression that controls flexibly for the parts of treatment exposure that are permanent or common across units. Wooldridge uses this perspective to connect TWFE to augmented pooled regressions.

2.5 Two-Way Mundlak

In two-way fixed effects models, we have both unit effects and time effects:

$$Y_{it} = X_{it}\beta + a_i + \lambda_t + u_{it}$$

The two-way Mundlak idea is to model both sources of correlation:

$$a_i = \bar{X}_i\pi_a + r_i$$

and

$$\lambda_t = \bar{X}_{.t}\pi_\lambda + s_t$$

where:

- $\bar{X}_{i.}$ is the unit-specific average of X ;
- $\bar{X}_{.t}$ is the time-specific average of X .

The resulting pooled regression includes:

$$Y_{it} = X_{it}\beta + \bar{X}_{i.}\pi_a + \bar{X}_{.t}\pi_\lambda + \text{residual components}$$

In a linear balanced panel, this is closely tied to the TWFE transformation:

$$X_{it} - \bar{X}_{i.} - \bar{X}_{.t} + \bar{X}_{..}$$

This is why Wooldridge emphasizes the equivalence between TWFE and the two-way Mundlak regression. It provides a way to understand TWFE as an augmented pooled regression rather than as a mysterious black-box demeaning operation.

That matters for DiD because it lets us add the interactions needed for heterogeneous effects while still keeping the familiar regression machinery.

3. Wooldridge's DiD Logic

3.1 The Problem With a Single TWFE Coefficient

In modern DiD, treatment effects often vary:

- across treated cohorts;
- across event time;
- across calendar time;
- across treatment intensity;
- across baseline characteristics.

If we estimate:

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + u_{it}$$

then β is forced to summarize all of that heterogeneity. In some designs, the resulting weighted average can be difficult to interpret.

Wooldridge’s solution is to write a more saturated regression. In a binary staggered-adoption setting, the target object is often a set of group-time average treatment effects:

$$ATT(g, t)$$

where g is the first treatment period and t is calendar time. A flexible regression can estimate these directly by interacting cohort indicators with time indicators.

3.2 Extended TWFE

The extended TWFE, or ETWFE, idea is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_g \sum_t \theta_{gt} 1\{G_i = g\} 1\{t\} + u_{it}$$

with appropriate omitted categories and sample restrictions. The coefficients θ_{gt} recover interpretable cohort-time treatment effects under conditional parallel trends.

The key difference from ordinary TWFE is that ETWFE does not force one common treatment coefficient. It lets treatment effects differ by cohort and time.

3.3 Conditional Parallel Trends

The identifying assumption is not simply “parallel trends” in the abstract. It is a restriction on untreated potential outcomes. A generic version is:

$$E(Y_{it}(0) - Y_{is}(0) | G_i = g, X_i) = E(Y_{it}(0) - Y_{is}(0) | G_i = \text{comparison}, X_i)$$

for relevant periods s and t .

In regression terms, this means we must specify the untreated outcome evolution richly enough. Wooldridge’s framework is attractive because it makes clear what is being conditioned on and how heterogeneity is handled.

3.4 Event Studies as Treatment-Effect Paths

The event-study version replaces a single post-treatment coefficient with relative-time coefficients:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \theta_k 1\{t - G_i = k\} + u_{it}$$

In Wooldridge-style implementations, the event-study coefficients should be constructed from well-defined group-time effects or from a regression that properly accounts for treatment-effect heterogeneity.

The pre-period coefficients are not proof of identification, but they are diagnostic. If high-exposure or early-treated units were already trending differently before treatment, the parallel-trends story is weaker.

3.5 Nonbinary and Continuous Treatments

The H-2A setting is not a binary treatment design. There is no clean “treated” versus “untreated” split. Instead, every county faces some AEW environment, and the relevant treatment is intensity:

$$Bite_{it} = AEW R_{r(i),t} - w_{cz(i),t}^{25}$$

In a continuous-treatment DiD, the target is usually a dose-response object:

$$\frac{\partial E[Y_{it}(d)|\cdot]}{\partial d}$$

or an average contrast between higher and lower treatment intensities:

$$E[Y_{it}(d_H) - Y_{it}(d_L)|\cdot]$$

The identifying assumption is a continuous-dose version of parallel trends. A simple version says that, absent the policy exposure, changes in outcomes would not have been systematically related to the treatment dose after conditioning on the chosen controls and fixed effects.

This is stronger than it may first appear. If the dose is partly determined by local wages, then treatment intensity may be correlated with local labor-market shocks that also affect H-2A demand.

4. Current H-2A Specification

4.1 Outcome

The main outcome is:

```
h2a_cert_share_farm_workers_2011_start_year
```

This is the number of H-2A workers certified in the start year, normalized by 2011 farm employment in the county. The normalization fixes the denominator at a pre-period baseline, which avoids mechanically changing the outcome because farm employment changes over time.

Let:

$$Y_{ct} = \frac{\text{H-2A workers certified}_{ct}}{\text{farm employment}_{c,2011}}$$

where c indexes counties and t indexes years.

4.2 Treatment Intensity

The main treatment variable is:

```
aewr_cz_p25_l1 = aewr_ppi_l1 - wage_p25_l1
```

In notation:

$$Bite_{ct-1} = AEW R_{r(c),t-1}^{real} - w_{cz(c),t-1}^{25}$$

where:

- $AEW R_{r(c),t-1}^{real}$ is the lagged real AEW in county c 's AEW region;
- $w_{cz(c),t-1}^{25}$ is the lagged 25th percentile agricultural wage in the county's commuting zone.

The treatment intensity is higher when the regional AEW is high relative to the local low-wage agricultural labor market.

This variable has two conceptually distinct components:

$$Bite_{ct-1} = \underbrace{AEW R_{r(c),t-1}^{real}}_{\text{regional policy wage}} - \underbrace{w_{cz(c),t-1}^{25}}_{\text{local wage distribution}}$$

This decomposition is central. The first component is set at the AEWR-region level. The second component is local and may reflect precisely the labor-market conditions that also affect H-2A usage.

4.3 Main Regression

The main regression is:

$$Y_{ct} = \alpha_c + \lambda_t + \beta_0 \text{Bite}_{ct-1} + \beta_1 (\text{Bite}_{ct-1} \times \text{Post}_t) + u_{ct}$$

with optional controls:

$$+ X'_{ct} \gamma$$

where:

- α_c are county fixed effects;
- λ_t are year fixed effects;
- $\text{Post}_t = 1$ after 2011;
- X_{ct} includes log population and the employment-to-population ratio in controlled specifications.

Because Post_t is collinear with year fixed effects, the model does not separately estimate a post dummy. The coefficient of interest is β_1 .

4.4 Interpretation of the Current Coefficient

The coefficient β_1 is a change in slope. It asks:

After 2011, did the relationship between lagged AEWR bite and H-2A usage become more positive or more negative, relative to the pre-2012 relationship, after removing county fixed effects and common year shocks?

It is not a simple treatment-control DiD. It is a continuous-dose DiD with a single post-period slope change.

The current event-study specification replaces Post_t with year indicators:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} \beta_\tau (\text{Bite}_{ct-1} \times 1\{t = \tau\}) + u_{ct}$$

The omitted year is 2011. Each β_τ is the slope relating bite to H-2A usage in year τ , relative to the slope in 2011.

This is closer to a Wooldridge-style event study because it does not collapse all post years into one coefficient. But it is still restrictive because it assumes a linear dose response and uses the time-varying bite itself as the dose.

5. Mapping Wooldridge to the H-2A Setting

5.1 What Are the “Groups”?

In canonical staggered DiD, the group is often the treatment cohort. In the H-2A setting, there is no treatment cohort because treatment does not switch from zero to one. We need to define groups in a way that makes sense for continuous exposure.

Possible group definitions include:

1. Baseline bite groups:

$$G_c = \text{quartile of } Bite_{c,2011}$$

2. Pre-period average bite groups:

$$G_c = \text{quartile of } \frac{1}{T_0} \sum_{t \leq 2011} Bite_{ct}$$

3. Baseline predicted bite groups, using only predetermined features:

$$G_c = \text{quartile of predicted AEWR exposure from pre-period crop/labor structure}$$

4. Continuous baseline dose:

$$D_c^0 = Bite_{c,2011}$$

The Wooldridge logic favors groups or doses that are predetermined. That is because a time-varying local wage component can be endogenous to contemporaneous labor-market shocks. A baseline bite measure is not automatically exogenous, but it is less vulnerable to post-period reverse causality.

5.2 A Wooldridge-Style Baseline-Dose Event Study

A natural specification is:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} \theta_{\tau}(D_c^0 \times 1\{t = \tau\}) + u_{ct}$$

where D_c^0 is a predetermined baseline dose, such as 2011 bite or pre-2012 average bite.

This asks whether counties with higher baseline exposure experience different post-2011 changes in H-2A usage, relative to counties with lower baseline exposure.

The pre-period coefficients are especially important:

$$\theta_{2008}, \theta_{2009}, \theta_{2010}$$

If these are nonzero or trending, then high-bite and low-bite counties were already on different trajectories before the relevant policy period.

This specification is easier to interpret than the current time-varying bite event study because the dose is fixed. It asks about differential trends by baseline exposure rather than continuously updating the dose using potentially endogenous local wage changes.

5.3 A Nonparametric Dose-Group Event Study

To avoid imposing a linear dose response, define bite groups:

```
samp_base <- samp_base %>%  
  group_by(county_fe) %>%  
  mutate(bite_2011 = aewr_cz_p25_l1[year == 2011]) %>%  
  ungroup() %>%  
  mutate(bite_2011_quartile = ntile(bite_2011, 4))
```

Then estimate:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{q \neq q_0} \sum_{\tau \neq 2011} \theta_{q\tau} 1\{G_c = q\} 1\{t = \tau\} + u_{ct}$$

In `fixest` style:

```

feols(
  h2a_cert_share_farm_workers_2011_start_year ~
    i(year, bite_2011_quartile, ref = 2011) |
    county_fe + year_fe,
  data = samp_base,
  vcov = ~cz_aewr_region_fe
)

```

This is often the most transparent descriptive version:

- it shows whether the effect is monotone in baseline bite;
- it avoids assuming linearity;
- it makes common-support problems visible;
- it produces an event-study plot by dose group.

The omitted group should be a substantively meaningful low-bite group, often the lowest quartile. The omitted year should remain 2011 if that is the base year for the current analysis.

5.4 A Rich Continuous-Dose Wooldridge Specification

If the goal is to preserve the continuous treatment interpretation, the model can be enriched:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} \theta_{\tau}(D_c^0 \times 1\{t = \tau\}) + \sum_{\tau \neq 2011} \eta_{\tau}(D_c^0)^2 \times 1\{t = \tau\} + u_{ct}$$

or by using splines:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} f_{\tau}(D_c^0) + u_{ct}$$

where f_{τ} is a flexible function of baseline bite.

This version is closer to the nonbinary-treatment Wooldridge logic. The dose is not forced into one linear slope. Instead, the dose-response relationship can vary over time.

5.5 Adding Covariate Heterogeneity

Wooldridge’s framework allows conditioning on covariates in a structured way. For this project, important baseline covariates might include:

- baseline farm employment;
- baseline crop acreage;
- crop mix;
- baseline H-2A use;
- predicted H-2A use;
- pre-period farm labor share;
- state minimum wage environment;
- region or crop-type groupings.

The point is not simply to add many controls. The point is to allow untreated trends to differ by baseline characteristics:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} \theta_{\tau}(D_c^0 \times 1\{t = \tau\}) + \sum_{\tau} X_c^0 \Gamma_{\tau} + u_{ct}$$

In `fixest`, this can be implemented using interactions between baseline covariates and year:

```
feols(  
  Y ~ i(year, baseline_bite, ref = 2011) +  
      i(year, baseline_farm_emp, ref = 2011) +  
      i(year, baseline_h2a_use, ref = 2011) |  
      county_fe + year_fe,  
  data = samp_base,  
  vcov = ~cz_aewr_region_fe  
)
```

This is a key Wooldridge-style move: if high-bite counties differ from low-bite counties in observable baseline ways, let those baseline differences predict different time paths. Then the identifying assumption is conditional on those differences.

5.6 Region-by-Year Fixed Effects as a Diagnostic

Because AEWR is set at the region-year level, an important diagnostic is:

$$Y_{ct} = \alpha_c + \lambda_{r(c),t} + \beta(Bite_{ct-1} \times Post_t) + u_{ct}$$

where $\lambda_{r(c),t}$ are AEWR-region-by-year fixed effects.

This absorbs all region-level AEWWR variation in a given year. What remains is within-region-year variation in the local wage component of the bite.

Interpretation:

- If the coefficient remains similar, the result is mostly driven by local wage-distribution variation within AEWWR regions.
- If the coefficient disappears, the result is mostly driven by AEWWR-region-year variation.
- If the coefficient changes sign, the AEWWR component and local wage component are pushing in different directions.

This is not necessarily a preferred causal specification, because it may remove the policy variation of interest. But it is a critical source-of-variation diagnostic.

6. How Mundlak Helps Interpret the H-2A Regressions

6.1 Current TWFE as Residualized Regression

The current main coefficient can be understood through Frisch-Waugh-Lovell. Define:

$$D_{ct} = Bite_{ct-1} \times Post_t$$

Because the regression also includes $Bite_{ct-1}$, county fixed effects, and year fixed effects, the identifying variation in D_{ct} is:

$$\tilde{D}_{ct} = D_{ct} - \hat{E}(D_{ct} | Bite_{ct-1}, \alpha_c, \lambda_t)$$

The coefficient is:

$$\hat{\beta}_1 = \frac{\sum_{ct} \tilde{D}_{ct} \tilde{Y}_{ct}}{\sum_{ct} \tilde{D}_{ct}^2}$$

where $\tilde{Y}_{ct}$ is the outcome residualized on the same controls except for D_{ct} .

This is the object whose distribution should be studied. Raw ECDFs of the bite are useful, but ECDFs of the residualized identifying regressor are more directly connected to the estimate.

6.2 The Mundlak Interpretation

A two-way Mundlak version of the same logic would include the relevant unit and time averages of the treatment variables rather than absorbing fixed effects directly. For a simplified treatment variable $Bite_{ct-1}$, the augmented pooled regression would include:

$$Bite_{ct-1}$$

plus county-level means:

$$\bar{Bite}_c.$$

and year-level means:

$$\bar{Bite}_t.$$

The coefficient on the time-varying bite is then interpreted as a within-county, net-of-year relationship, not as a cross-sectional comparison between high-bite and low-bite counties.

For the H-2A paper, this helps communicate the design:

We are not comparing permanently high-bite counties to permanently low-bite counties. County fixed effects, or equivalently the Mundlak county averages, absorb permanent differences in exposure. Year fixed effects, or equivalently time averages, absorb common shocks. The identifying variation is the residual movement in bite and bite-by-post after those components are removed.

That statement is helpful, but it also exposes the key weakness:

If the residual movement in bite is driven by local wage shocks that also directly affect H-2A demand, the fixed effects/Mundlak structure does not solve the endogeneity problem.

7. Recommended Wooldridge-Style Specifications for the Project

7.1 Specification A: Current Model, Rewritten as a Dose DiD

Keep the current model, but describe it explicitly:

$$Y_{ct} = \alpha_c + \lambda_t + \beta_0 Bite_{ct-1} + \beta_1 Bite_{ct-1} Post_t + X'_{ct} \gamma + u_{ct}$$

Interpret β_1 as:

the post-2011 change in the slope of H-2A usage with respect to lagged AEWR bite.

This should be the baseline, not the final word.

7.2 Specification B: Flexible Year-by-Bite Event Study

This is already close to what the code does:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} \beta_\tau \text{Bite}_{ct-1} 1\{t = \tau\} + u_{ct}$$

This is useful because it shows when the relationship emerges. In the current results, the negative coefficients become more pronounced in later years, especially after the mid-2010s.

The limitation is that Bite_{ct-1} remains time-varying and includes local wage variation.

7.3 Specification C: Baseline-Bite Wooldridge Event Study

Use a predetermined dose:

$$D_c^0 = \text{Bite}_{c,2011}$$

or:

$$D_c^0 = \frac{1}{4} \sum_{t=2008}^{2011} \text{Bite}_{ct}$$

Then estimate:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} \theta_\tau D_c^0 1\{t = \tau\} + u_{ct}$$

This is probably the cleanest Wooldridge-style continuous-dose event study for the paper. It defines treatment intensity before the post period and then asks whether higher-exposure counties diverge afterward.

7.4 Specification D: Baseline-Bite Quartile Event Study

Define exposure groups using pre-period bite:

$$G_c \in \{Q1, Q2, Q3, Q4\}$$

Estimate:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{q=2}^4 \sum_{\tau \neq 2011} \theta_{q\tau} 1\{G_c = q\} 1\{t = \tau\} + u_{ct}$$

This provides the most visually intuitive robustness check:

- low-bite counties are the reference;
- higher-bite groups trace out differential event-study paths;
- pre-period differences are easy to inspect;
- nonlinearities are visible.

This is especially useful if the continuous-dose coefficient is driven by tails of the bite distribution.

7.5 Specification E: Conditional Wooldridge Event Study

Add baseline covariates interacted with year:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} \theta_{\tau} D_c^0 1\{t = \tau\} + \sum_{\tau} X_c^0 \Gamma_{\tau} + u_{ct}$$

This is a richer conditional trends model. It says high- and low-bite counties may differ in baseline characteristics, and those characteristics may predict different trends. Identification then comes from differences in bite conditional on those trend predictors.

This is closer to Wooldridge's favored approach than simply adding contemporaneous controls.

8. Source-of-Variation Diagnostics

Before choosing the final Wooldridge-style specification, the paper should show where the estimate comes from.

8.1 Decompose the Bite

Report variation in:

$$AEWR_{r(c),t-1}^{real}$$

and:

$$w_{cz(c),t-1}^{25}$$

separately.

Useful diagnostics:

1. Raw variance of each component.
2. Within-county, net-of-year variance of each component.
3. Correlation of residualized bite with residualized AEWR.
4. Correlation of residualized bite with residualized local p25 wage.
5. Share of residualized bite variance explained by each component.

If the residualized bite is mostly local-wage variation, the identifying story is not mainly about AEWR policy variation. If it is mostly regional AEWR variation, then aggregate inference and region-level shocks become central.

8.2 ECDFs and Quantile Plots

ECDFs are useful, but they should be used for several objects:

1. Raw bite by year.
2. Raw bite by baseline bite group.
3. Residualized bite after county and year fixed effects.
4. Residualized interaction term:

$$Bite_{ct-1} \widetilde{Post}_t$$

5. Outcome residuals by exposure group.

The current script constructs an ECDF-like plot using the demeaned outcome and treatment groups. That is a good start, but the more direct Wooldridge diagnostic is the distribution of the residualized identifying regressor and its covariance with the residualized outcome.

8.3 Leverage and Contribution Diagnostics

For the main coefficient, define the residualized regressor:

```
D <- samp_base$aewr_cz_p25_l1 * samp_base$postdummy
```

Residualize D on:

- aewr_cz_p25_l1;
- county fixed effects;
- year fixed effects.

Then compute:

$$Leverage_{ct} = \frac{\tilde{D}_{ct}^2}{\sum_{c,t} \tilde{D}_{ct}^2}$$

and:

$$Contribution_{ct} = \frac{\tilde{D}_{ct} \tilde{Y}_{ct}}{\sum_{c,t} \tilde{D}_{ct} \tilde{Y}_{ct}}$$

Aggregate these by:

- year;
- AEW region;
- CZ;
- baseline bite quartile;
- crop region or crop specialization;
- border versus non-border CZs.

This answers the practical question:

Which observations are actually driving the coefficient?

8.4 Leave-One-Out Sensitivity

At minimum, report:

- leave-one-AEWR-region-out estimates;
- leave-one-year-out estimates;
- leave-one-CZ-region-cluster-out estimates for influential clusters;
- estimates excluding the largest farm-employment CZs within AEWR regions.

The current code already begins this logic by dropping large CZs within AEWR regions. A Wooldridge-style diagnostic framework would make this systematic and connect it to the identifying variation.

8.5 Support and Overlap

For continuous treatment, overlap means that the comparison units have meaningful support across the dose distribution. Useful checks:

- pre-period support of bite by post-period exposure group;
- support of baseline covariates by bite quartile;
- whether high-bite counties are concentrated in a few AEWR regions;
- whether the upper tail of bite is concentrated in a few crop systems.

If the highest bite quartile is mostly one or two regions or one crop system, then a national continuous-dose coefficient may be too broad.

9. Inference Implications

The current main regressions cluster at:

```
cz_aewr_region_fe
```

This gives many clusters and is appropriate for CZ-level serial correlation and for CZs that cross AEWR region boundaries. But the AEWR component of the treatment varies at the AEWR-region-year level. This creates a Moulton-style concern: observations within the same AEWR region and year share part of the treatment.

For the current bite variable, the appropriate clustering level is not obvious because the treatment combines:

- regional AEWR variation;
- local CZ wage variation.

Recommended practice:

1. Report the current CZ-by-AEWR-region clustered standard errors.
2. Report state-clustered standard errors.
3. Report AEWR-region-clustered standard errors as a conservative check.
4. For IV specifications using region-year instruments, use AEWR-region clustering and wild cluster bootstrap methods.

If the coefficient is sensitive to moving from CZ-region clustering to AEWR region clustering, the paper should say so directly. That sensitivity is not a minor technicality; it reveals that aggregate policy variation is part of the identifying design.

10. How the IV Robustness Checks Fit With Wooldridge

The Wooldridge approach and the IV robustness checks answer different questions.

The Wooldridge-style regression answers:

Can we specify the DiD comparison flexibly enough that the dose-response relationship is transparent and credible under conditional parallel trends?

The IV strategies answer:

Can we isolate a component of AEWR bite that is plausibly unrelated to local shocks affecting H-2A usage?

10.1 Measurement-Error Revision IV

The FLS preliminary-to-final revision instrument targets administrative measurement error in the AEWR-setting process:

$$Z_{rt} = \hat{W}_{r,t-1}^{Nov} - W_{r,t-1}^{final}$$

This is attractive because it isolates variation in the AEWR that comes from survey revision noise, not from local agricultural labor demand. Conceptually, this is a strong complement to Wooldridge-style DiD:

- Wooldridge handles heterogeneity and transparent aggregation.
- The revision IV targets endogeneity in the treatment intensity.

The main risks are weak first-stage power and few-cluster inference.

10.2 Agro-Dissimilar Leave-Cluster-Out IV

The agro-dissimilar instrument uses wage movements in other agricultural subregions inside the same FLS region. It is likely more powerful, but the exclusion restriction is harder:

- other subregions may share commodity shocks;
- farm labor contractors may operate across subregions;
- migrant labor flows may link the markets;
- weather and input cost shocks may be regionally correlated;
- clustering choices create researcher degrees of freedom.

This design can still be useful, but it should be presented as a sensitivity analysis with validation exercises rather than as the cleanest identification strategy.

11. Recommended Empirical Roadmap

Step 1: Keep the Current Model as Baseline

Present:

$$Y_{ct} = \alpha_c + \lambda_t + \beta_0 \text{Bite}_{ct-1} + \beta_1 \text{Bite}_{ct-1} \text{Post}_t + u_{ct}$$

Describe it as a continuous-dose DiD, not as a standard treated-control DiD.

Step 2: Add the Current Flexible Event Study

Present:

$$Y_{ct} = \alpha_c + \lambda_t + \sum_{\tau \neq 2011} \beta_\tau \text{Bite}_{ct-1} 1\{t = \tau\} + u_{ct}$$

Use it to show pretrends and the timing of post-period divergence.

Step 3: Add Baseline-Dose Wooldridge Event Studies

Estimate both:

$$\sum_{\tau \neq 2011} \theta_{\tau} D_c^0 1\{t = \tau\}$$

and the quartile version:

$$\sum_q \sum_{\tau \neq 2011} \theta_{q\tau} 1\{G_c = q\} 1\{t = \tau\}$$

These should be the central robustness checks for the continuous-treatment DiD.

Step 4: Add Conditional Trend Controls

Interact baseline covariates with year. Start with a small, defensible set:

- baseline farm employment;
- baseline H-2A use or predicted H-2A use;
- baseline crop acreage;
- baseline crop mix or crop-type shares;
- baseline population or labor-market size.

The goal is not to maximize controls. The goal is to show that the result is not just high-bite counties following different predictable paths.

Step 5: Run Source-of-Variation Diagnostics

Report:

- residualized treatment ECDFs;
- leverage by year and AEWR region;
- numerator contribution by year and AEWR region;
- leave-one-region-out estimates;
- AEWR component versus local wage component decomposition.

These diagnostics are essential for a credible Wooldridge-style presentation.

Step 6: Use IV as a Separate Robustness Design

If the measurement-error revision instrument has power, present it as the preferred IV robustness check. Use the agro-dissimilar instrument as a secondary check if the validation exercises support it.

12. Suggested Language for the Paper

The current design can be described as follows:

Our baseline specification is a continuous-treatment difference-in-differences design. Rather than comparing treated and untreated counties, we compare counties with different intensities of exposure to the AEWR bite, defined as the gap between the lagged real AEWR and the lagged local agricultural wage distribution. County fixed effects absorb permanent differences across counties, and year fixed effects absorb shocks common to all counties. The coefficient on the interaction between bite and the post-2011 indicator captures the post-period change in the slope of H-2A usage with respect to the AEWR bite.

A Wooldridge-style extension can be described as:

Following Wooldridge’s regression-based approach to modern DiD, we also estimate flexible dose-response event-study specifications. These models replace the single post-period slope with year-specific exposure effects and, in some specifications, define exposure using pre-period bite groups. This allows us to inspect pretrends, relax the assumption of a constant post-period effect, and assess whether the results are driven by a narrow portion of the exposure distribution.

The limitation should be stated clearly:

Because the AEWR bite combines regional AEWR variation with local wage variation, the dose may be correlated with local labor-market shocks. The Wooldridge-style specifications improve transparency and allow richer heterogeneity, but they do not by themselves eliminate this source of endogeneity. We therefore complement them with instrumental-variable robustness checks that isolate plausibly exogenous components of AEWR variation.

13. Main Takeaways for the H-2A Project

1. Wooldridge’s approach is not a single estimator. It is a way to write DiD regressions so that heterogeneity, timing, conditioning variables, and aggregation are explicit.

2. Mundlak-Chamberlain is the bridge between fixed effects and augmented pooled regression. It shows that fixed effects identification can be understood as controlling for unit-level and time-level averages of the treatment variables.
3. For this project, the most useful Wooldridge move is to replace the single post-period dose coefficient with flexible year-by-dose and dose-group event studies.
4. The key empirical issue is the source of treatment variation. The bite is a difference between a regional policy wage and a local wage percentile. A credible empirical section should show how much of the result is driven by each component.
5. Baseline or pre-period bite is likely a more credible dose variable for the Wooldridge-style DiD robustness check than fully time-varying bite, because it avoids mechanically updating treatment intensity with post-period local wage shocks.
6. Region-by-year fixed effects are a diagnostic, not necessarily the preferred model. They reveal whether the result is driven by regional AEWR variation or within-region local wage variation.
7. The measurement-error revision IV is conceptually the cleaner IV robustness check if it has enough first-stage power. The agro-dissimilar leave-cluster-out IV is potentially useful but rests on a more demanding exclusion restriction.